Agricore Farm System

FFMS: one platform for crops, livestock, resources and money

with Laravel + React

one connected system

Foundation

Intro, architecture, roles, dashboard

Field

Crops, land, irrigation, protection

Infrastructure

Livestock, equipment, storage, tasks

Business

Finance, suppliers, sales, future

The problem and our goal

1

The problem

Traditional farming runs on paper, memory and guesswork, so time and money leak away.

2

Our goal

One web platform that manages every part of the farm in a single place.

3

Who it is for

Farm owners, managers and workers who need clear, shared records.

4

What it delivers

Better decisions, fewer losses and a full picture of farm performance.

System architecture



The frontend talks to the backend through a JSON API.

Core setup: roles and the Farm

Admin

Oversees the system

Farm Owner

Controls the farm and assigns staff



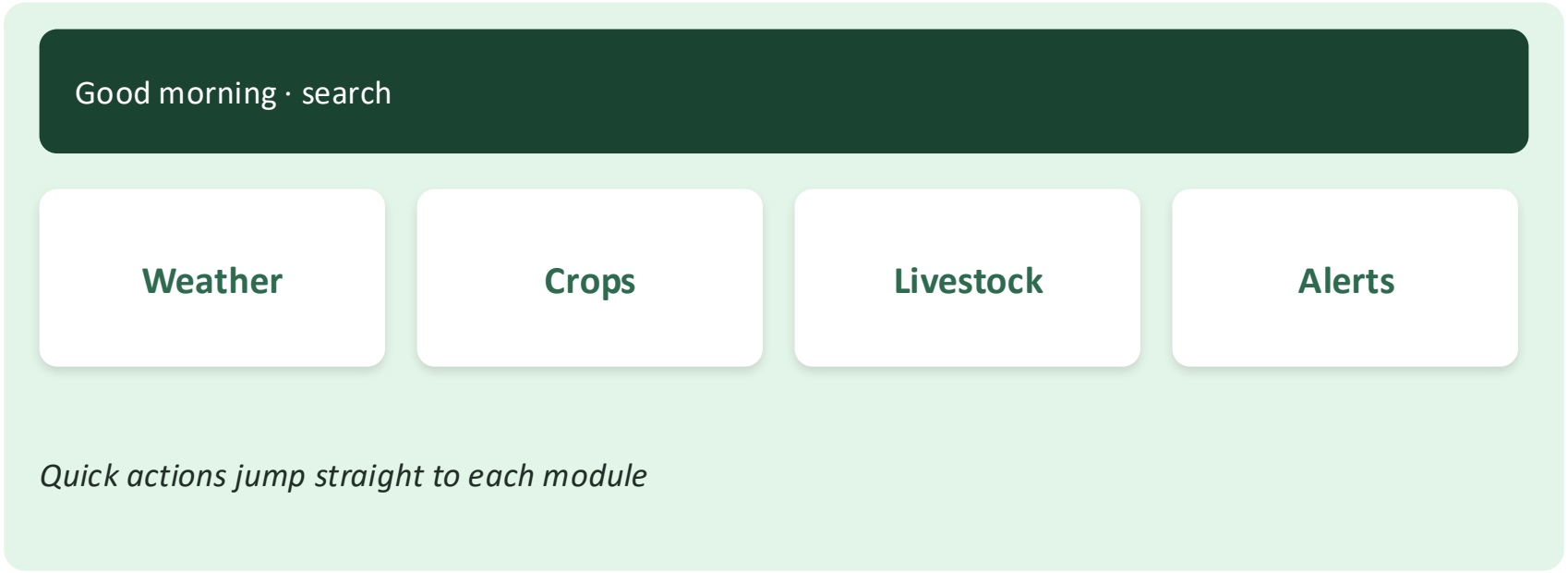
Employee

Does the daily work

Everything else

Crops, livestock, stock, finance

Dashboard overview



Crops, fields and land

1

Crop lifecycle

Planting schedules, crop production and yield records.

2

Field & land

Field boundaries, mapping and resource allocation.

From seed to harvest, every field has a history.

Precision farming and protection

1

Irrigation

Schedules and pump monitoring so water is used wisely.

2

Weather

Live data through WeatherService guides decisions.

3

Scouting

Field scouting records catch problems early.

4

Spraying & alerts

Spraying schedules and outbreak alerts protect the crop.

Livestock and equipment

1

Livestock

Breeding records, health tracking and vaccination schedules.

2

Equipment

Maintenance records, fuel logs and equipment inventory.

Healthy animals and working machines keep the farm moving.

Storage, logistics and daily operations

1

Post-harvest

Handling produce correctly after it leaves the field.

2

Storage

Facility management and stored-produce records.

3

Daily tasks

A task system that keeps the day organised.

4

Assignments

Employees get clear work, owners see progress.

Finance and supply chain

1

Financials

Budgeting, expense tracking and transaction reports.

2

Quotations

Compare supplier quotations before buying.

3

Purchase orders

Create, approve or reject orders.

4

Payments

Procurement payments tracked end to end.

Sales and market

1. **Market prices**
   Track prices to pick the right time to sell.

2. **Sales orders**
   Record every order from start to finish.

3. **Invoices**
   Bill customers clearly and on time.

4. **Customers**
   Keep buyer details and history in one place.

Conclusion and future scope

What we achieved and where next

Achieved

- One connected platform for the whole farm
- Role-based access around a shared Farm entity
- Modules covering crops to cash

Next

- AI-driven yield prediction
- Smarter price and demand insight
- Live hosted deployment